Zaccharie Ramzi | AI Scientist

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Researcher specialized in Deep Learning with excellent Technical Skills and an appetite for Research

Professional experiences

- **Mistral AI** — **Paris**
  AI Scientist — *2025-*
  Working in the alignment team

- **Meta - Reality Labs** — **Paris**
  AI Research Scientist — *2023-2025*
  Research on EMG for neural interfaces

- **CNRS - ENS** — **Paris**
  Postdoctoral researcher — *2022-2023*
  In the team of Gabriel Peyré:
  - Research on optimization, implicit models and unrolled models
  - Participation in several open source projects, including their respective sprints: benchopt, jaxopt, ott-jax
  - Fostering the lab's life: scientific seminars organization, creation of a wiki

- **CEA & Inria** — **Paris**
  PhD Student — *2019-2022*
  In the Parietal and Cosmostat teams, my PhD was focused on Deep Learning for MRI reconstruction.
  - Designed and implemented new models for MRI reconstruction which allowed me to secure the **2nd spot in the fastMRI 2020 reconstruction challenge** organized by Facebook and NYU. This was featured in 2 specialized press articles (CEA mag and Contact).
  - Used the public HPC Jean Zay to train neural networks in a distributed fashion with up to 8 nodes totaling 32 GPUs while working on 1Tb of MRI data. Co-created a user's collaborative documentation.
  - Co-founded a lecture group focused on Deep Learning for students at NeuroSpin.
  - Analyzed and benchmarked the state-of-the-art in Deep Learning for MRI reconstruction.
  - Published 18 papers/abstracts in international peer-reviewed conferences/workshops/journals, among which **IEEE TMI** and **ICLR** (spotlight).
  - Co-supervised 2 interns, totaling 8 months.
  - Peer-reviewed 19 submissions for scientific conferences and journals.

- **xbird** — **Berlin**
  Data scientist — *2017-2019*
  In order to passively monitor diabetes patients, I built data pipelines, read scientific literature, specifically on human activity recognition, designed and implemented ad hoc machine learning models to efficiently detect human activities with smartphone and wearable data (GPS, accelerometer), deployed to production and maintained said models, presented research results to both tech and non-tech teams.

- **BioSerenity - Brain and Spine Institute** — **Paris**
  Research intern — *April–August 2017*

- **Celmatix** — **New-York**
  Data engineer and data science intern — *March–September 2016*

- **Ekimetrics** — **Paris**
  Data science intern — *August 2015–February 2016*

Education

- **CEA NeuroSpin & Cosmostat - INRIA Saclay Parietal team, PhD** **Paris**
  PhD in deep learning for image reconstruction *Since February 2019*
  The main focus of my thesis is to build an architecture that allows the reconstruction of undersampled MRI data. This led me to work in the fields of compressed sensing, optimization, computer vision, deep learning, and wavelets.
- **ENS Cachan, MVA** **Cachan**
  MSc in Mathematics, Vision and Learning, graduated with highest honors *2016–2017*
  Courses in statistical learning, optimization, graphs, graphical models, reinforcement learning, object recognition, kernel methods, text mining, bandit theory
- **Telecom ParisTech** **Paris**
  Engineering diploma, specialization in machine learning *2013–2017*
  Courses in mathematical statistics, optimization, relational databases, data mining, data visualization, introduction to machine learning

Technical skills

- **My go-to programming language:** Python (TensorFlow, PyTorch, Numpy, Pytest, Hydra, W&B, Jax, ...)
- **My typical development environment:** VS Code + Git + Jupyter notebooks
- **Programming languages I have used:** Java, Scala, Ruby, Javascript, R, Matlab, Shell
- **Other tools in my stack:** Docker, SLURM, LaTex, Linux, GitHub CI, Travis CI, PyPI
- **Open source contributions:** coremltools, TensorFlow, scikit-image, jaxopt, keras-cv

Open source projects

A selection of the open sources projects I take part in. All of them are on Github.

- **fastmri-reproducible-benchmark**
  Deep Learning for MRI reconstruction `fastmri-reproducible-benchmark`
  Using the code in this repo, I secured the 2nd spot in the fastMRI 2020 brain reconstruction challenge.
- **Benchopt**
  Benchmarking of optimization algorithms `benchopt`,`benchmark_resnet_classif`
  A meta-benchmarking tools allowing to build benchmarks for optimization algorithms in all application settings.
- **jaxopt**
  Optimization in jax `jaxopt`
  Hardware accelerated, batchable and differentiable optimizers in JAX.

Miscellaneous

- I was a private tutor from 2012 until 2015. I helped high school students in maths and physics. I am now a volunteer tutor since 2019 for Entraide Scolaire Amicale.
- French is my mother tongue and I speak English at a C1 level (TOEFL: 110). I can speak a bit of conversational German.
- I was part of a football team for 5 years, mostly during my high school years. In the process we secured the first spot in the second departmental division and got promoted to the first division.
- I am a huge Coen Brothers and Stanley Kubrick enthusiast. More generally I love cinema, and enjoy a lot watching the web-serie *Blow up* by ARTE.

Publications

G R, C., **Ramzi, Z.**, Ciuciu, P., "Hybrid learning of Non-Cartesian k-space trajectory and MR image reconstruction networks," Accepted as an oral to ISBI 2022, 2022, [Online]. Available: https://hal.inria.fr/hal-03394881.

Ramzi, Z., Starck, J.-L., G R, C., Ciuciu, P., "NC-PDNet: a Density-Compensated Unrolled Network for 2D and 3D non-Cartesian MRI Reconstruction," Accepted for publication at IEEE Transactions on Medical Imaging, 2021, [Online]. Available: https://hal.inria.fr/hal-03188997.

Farrens, S., Grigis, A., El Gueddari, L., **Ramzi, Z.**, G R, C., Starck, S., Sarthou, B., Cherkaoui, H., Ciuciu, P., Starck, J.-L., "PySAP: Python Sparse Data Analysis Package for multidisciplinary image processing," *Astronomy and Computing*, vol. 32, 2020, ISSN: 22131337. DOI: 10.1016/j.ascom.2020.100402. [Online]. Available: https://arxiv.org/abs/1910.08465.

Ramzi, Z., Remy, B., Lanusse, F., Starck, J.-L., Ciuciu, P., "Denoising Score-Matching for Uncertainty Quantification in Inverse Problems," in *NeurIPS 2020 Deep Learning and Inverse Problems workshop*, 2020. arXiv: 2011.08698. [Online]. Available: http://arxiv.org/abs/2011.08698.

Ramzi, Z., Ciuciu, P., Starck, J.-L., "Benchmarking proximal methods acceleration enhancements for CS-acquired MR image analysis reconstruction," in *SPARS 2019 - Signal Processing with Adaptive Sparse Structured Representations Workshop*, 2019. arXiv: 1807.04005. [Online]. Available: https://hal.inria.fr/hal-02298569.

——, "Benchmarking Deep Nets MRI Reconstruction Models on the FastMRI Publicly Available Dataset," in *ISBI 2020 - International Symposium on Biomedical Imaging*, 2020. [Online]. Available: https://hal.inria.fr/hal-02436223.

——, "Density Compensated Unrolled Networks for Non-Cartesian MRI Reconstruction," in *2021 IEEE 18th International Symposium on Biomedical Imaging (ISBI)*, 2021. arXiv: 2101.01570. [Online]. Available: http://arxiv.org/abs/2101.01570.

——, "Benchmarking MRI reconstruction neural networks on large public datasets," *Applied Sciences (Switzerland)*, vol. 10, no. 5, 2020, ISSN: 20763417. DOI: 10.3390/app10051816. [Online]. Available: https://www.mdpi.com/2076-3417/10/5/1816.

Ramzi, Z., Mannel, F., Bai, S., Starck, J.-L., Ciuciu, P., Moreau, T., "SHINE: SHaring the INverse Estimate from the forward pass for bi-level optimization and implicit models," in *ICLR*, **Spotlight**, 2022.

Remy, B., Lanusse, F., **Ramzi, Z.**, Liu, J., Jeffrey, N., Starck, J.-L., "Probabilistic Mapping of Dark Matter by Neural Score Matching," in *NeurIPS 2020 Machine Learning for Physical sciences workshop*, 2020, pp. 1–6. arXiv: 2011.08271. [Online]. Available: http://arxiv.org/abs/2011.08271.

G R, C., **Ramzi, Z.**, Ciuciu, P., "Learning the sampling density in 2D SPARKLING MRI acquisition for optimized image reconstruction," To be presented at Eusipco 2021, 2021, [Online]. Available: https://hal.inria.fr/hal-03158831v2.

Riemenschneider, B., Muckley, M., Radmanesh, A., Kim, S., Jeong, G., Ko, J., Jun, Y., Shin, H., Hwang, D., Mostapha, M., Arberet, S., Nickel, D., **Ramzi, Z.**, Ciuciu, P., Starck, J. L., Teuwen, J., Karkalousos, D., Zhang, C., Sriram, A., Huang, Z., Yakubova, N., Lui, Y. W., Knoll, F., "Results of the 2020 fastMRI Brain Reconstruction Challenge," in *ISMRM*, **Oral**, 2021. arXiv: 1811.08839. [Online]. Available: http://arxiv.org/abs/1811.08839.

Ramzi, Z., Ciuciu, P., Starck, J.-L., "XPDNet for MRI Reconstruction: an application to the 2020 fastMRI challenge," in *ISMRM*, **Oral**, 2021, pp. 1–4. arXiv: 2010.07290. [Online]. Available: http://arxiv.org/abs/2010.07290.

Ramzi, Z., Vignaud, A., Starck, J.-L., Ciuciu, P., "Is good old GRAPPA dead?" In *ISMRM*, 2021, ISBN: 2106.00753v1. arXiv: 2106.00753v1. [Online]. Available: https://arxiv.org/abs/2106.00753.

Muckley, M. J., Riemenschneider, B., Radmanesh, A., Kim, S., Jeong, G., Ko, J., Jun, Y., Shin, H., Hwang, D., Mostapha, M., Arberet, S., Nickel, D., **Ramzi, Z.**, Ciuciu, P., Starck, J. L., Teuwen, J., Karkalousos, D., Zhang, C., Sriram, A., Huang, Z., Yakubova, N., Lui, Y. W., Knoll, F., "Results of the 2020 fastMRI Challenge for Machine Learning MR Image Reconstruction," *IEEE Transactions on Medical Imaging*, pp. 1–12, 2021, ISSN: 1558254X. DOI: 10.1109/TMI.2021.3075856. [Online]. Available: https://arxiv.org/abs/2012.06318.

Ramzi, Z., Starck, J.-L., Moreau, T., Ciuciu, P., "Wavelets in the deep learning era," in *European Signal Processing Conference*, **Oral**, 2020, pp. 1417–1421, ISBN: 9789082797053. [Online]. Available: https://hal.archives-ouvertes.fr/hal-03020214.

Moreau, T., Massias, M., Gramfort, A., Ablin, P., Bannier, P.-A., Charlier, B., Dagréou, M., Tour, T. D., Durif, G., Dantas, C. F., Klopfenstein, Q., Larsson, J., Lai, E., Lefort, T., Malézieux, B., Moufad, B., Nguyen, B. T., Rakotomamonjy, A., **Ramzi, Z.**, Salmon, J., Vaiter, S., "Benchopt: Reproducible, efficient and collaborative optimization benchmarks," Jun. 2022.

Ramzi, Z., Michalewicz, K., Starck, J.-L., Moreau, T., Ciuciu, P., "Wavelets in the deep learning era," *Journal of Mathematical Imaging and Vision*, vol. 65, pp. 240–251, Oct. 2022.

Ramzi, Z., Ablin, P., Peyré, G., Moreau, T., "Test like you train in implicit deep learning," *ArXiv preprint arXiv:2305.15042*, 2023. [Online]. Available: https://arxiv.org/abs/2012.06318.